

Introduction

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This book presents the current state of the T0 time-mass duality framework and its applications to particle masses, fundamental constants, quantum mechanics, gravitation, and cosmology.

The main body of the book consists of a series of core documents on T0. These chapters reflect the present understanding of the theory and its quantitative consequences. Wherever possible, the material has been reorganized and unified to make the structure of the theory as transparent as possible.

At the end of the book, several older documents are included in an appendix. These texts represent earlier stages in the development of the T0 framework. They have not been removed because they make visible the evolution of the ideas and the refinement of the formulas. In many cases, one can observe how approximations were improved, special cases generalized, and how new empirical data helped sharpen or correct earlier arguments.

The "live" version of the theory is maintained in a public GitHub repository:

<https://github.com/jpascher/T0-Time-Mass-Duality>

The LaTeX sources of the chapters in this book originate from this repository. When conceptual or numerical errors are found, they are corrected there first. This means that the PDF version of the book you are reading is a snapshot of a continuously evolving project. For the most current version of the documents, including new appendices or corrections, the GitHub repository should always be considered the primary reference.

The intent of this compilation is twofold:

- to provide a coherent, readable path through the core ideas and results of the T0 framework;
- to document the historical development of these ideas in the appendix, including false starts, intermediate formulations, and early adjustments to experimental data.

Readers primarily interested in the current formulation of the theory may focus on the core chapters. Readers also interested in the thinking and trial-and-error process behind the theory are invited to study the appendix material in parallel.